Introduction to Microcontrollers

Arduino, ESP32, and Raspberry Pi — Architecture and Comparison

Arduino Uno **ESP32 Family** **Raspberry Pi**

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OVERVIEW

Learning Objectives

01 Explain what a microcontroller is and how it differs from a microprocessor / SBC.

02 Identify the major Arduino board types and when to use each.

03 Describe Arduino Uno's architecture: CPU, RAM, ROM/Flash, I/O.

04 Compare ESP32 variants (S2, S3, C3, C6) by cores, RAM, and connectivity.

05 Explain the Raspberry Pi family and how an SBC differs from a microcontroller board.

06 Select the right board (Arduino / ESP32 / Raspberry Pi) for a given robotics or IoT task.

FOUNDATIONS

What is a Microcontroller?

A microcontroller (MCU) is a single chip that integrates a processor core, memory, and input/output peripherals — a complete small computer designed to run one dedicated task inside a larger device.

- **CPU Core** — Executes program instructions (e.g., ATmega328P, Xtensa LX6/LX7, RISC-V).
- **RAM (SRAM)** — Volatile working memory for variables while the program runs — lost on power-off.
- **ROM / Flash** — Non-volatile storage that holds the compiled program (your sketch) permanently.
- **I/O Pins** — Digital and analog pins used to sense and control external circuits.

MCU vs. MPU

Microcontroller (MCU)

CPU + RAM + ROM on one chip; runs a single program (Arduino, ESP32).

Microprocessor (MPU)

CPU only; needs external RAM, storage, and an OS (used in a PC / Raspberry Pi's SoC).

Rule of thumb: if it boots an operating system, it's a microprocessor-based board (like Raspberry Pi). If it just runs your uploaded code, it's a microcontroller (like Arduino or ESP32).

ARDUINO FAMILY

Types of Arduino Boards

Uno

Standard learning board. ATmega328P, most tutorials & shields.

Nano

Breadboard-friendly, same core as Uno, much smaller footprint.

Mega 2560

54 digital I/O, 4 UARTs, large flash — for complex, pin-heavy projects.

Leonardo / Micro

ATmega32u4 with native USB — can act as a keyboard/mouse (HID).

Due

32-bit ARM Cortex-M3, 3.3V logic — higher speed, more memory.

Nano 33 IoT / UNO R4 WiFi

Built-in Wi-Fi/BLE for IoT projects, cloud-connected.

Most Popular Arduino Boards



ARDUINO NANO



MEGA 2560 REV 3



ARDUINO PRO MINI



ARDUINO UNO R3



MKR WAN 1300/1310



ARDUINO ETHERNET SHIELD 2



ARDUINO MICRO



ARDUINO DUE



NANO RP2040



UNO R4 WiFi

ARDUINO FAMILY

Arduino Uno — Architecture in Detail

Microcontroller	ATmega328P (8-bit AVR)
Clock Speed	16 MHz
Flash (ROM)	32 KB (~0.5 KB used by bootloader)
SRAM (RAM)	2 KB
EEPROM	1 KB (non-volatile, for saved data)
Digital I/O Pins	14 (6 support PWM: 3, 5, 6, 9, 10, 11)
Analog Input Pins	6 (A0–A5), 10-bit ADC (0–1023)
Operating Voltage	5 V logic

Why it matters

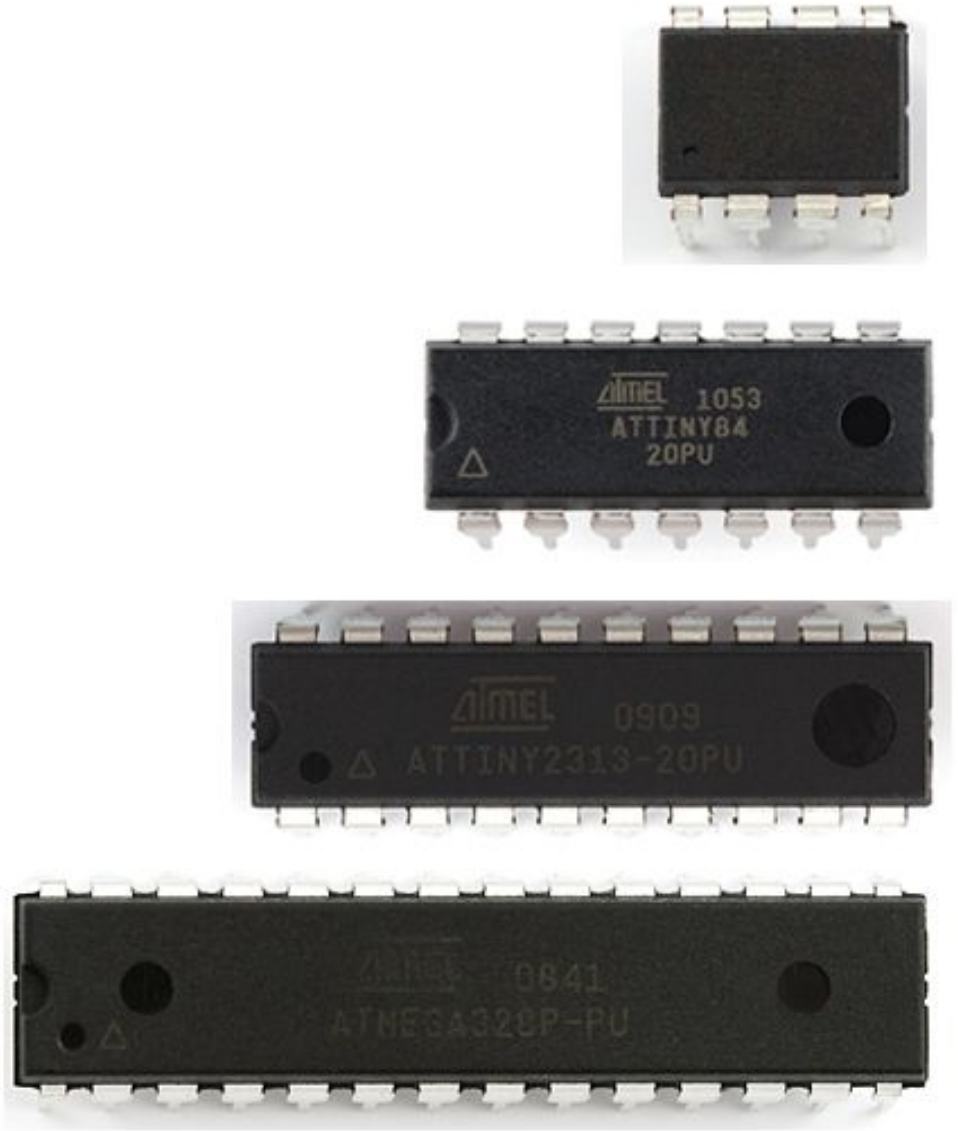
2 KB of RAM is tiny — every variable, array, and Serial string competes for that space. This is why large programs on Uno can compile fine but crash at runtime: SRAM, not flash, is usually the limiting resource.

PWM pins (~) simulate analog output by rapidly switching digital HIGH/LOW — the basis for LED dimming and motor speed control.

Arduino Board Comparison

Board	MCU	Clock	Flash	SRAM	Digital I/O	USB
Uno	ATmega328P	16 MHz	32 KB	2 KB	14	Standard
Nano	ATmega328P	16 MHz	32 KB	2 KB	14 (+2 analog)	Mini/Micro
Mega 2560	ATmega2560	16 MHz	256 KB	8 KB	54	Standard
Leonardo	ATmega32u4	16 MHz	32 KB	2.5 KB	20	Native (HID)
Due	ARM Cortex-M3	84 MHz	512 KB	96 KB	54	Native

Takeaway: memory (Flash & SRAM), not clock speed, is usually the deciding factor between these boards.


1053
ATTINY84
20PU
0909
ATTINY2313-20PU
0841
ATMEGA328P-PU

ESP32 FAMILY

Introduction to ESP32

ESP32 is a family of low-cost, low-power System-on-Chip (SoC) microcontrollers from Espressif, with built-in Wi-Fi and Bluetooth — designed to bring wireless IoT connectivity to embedded projects that Arduino Uno cannot provide on its own.

2
CPU Cores (dual-core Xtensa LX6, classic ESP32)

520 KB
SRAM — 260x more than Arduino Uno

240 MHz
Max clock speed — 15x faster than Uno

Typical Uses

- Wi-Fi / Bluetooth IoT sensors
- Home automation & smart devices
- WiFi-based human activity / CSI sensing
- Wearables and low-power BLE beacons
- Camera & edge-AI modules (S3)

ESP32 FAMILY

ESP32 Variants Compared

Variant	Cores / Arch	Max Clock	SRAM	Wi-Fi	Bluetooth	Notable Feature
ESP32 (classic)	Dual-core Xtensa LX6	240 MHz	520 KB	Yes	Classic + BLE	Most widely used, richest peripherals
ESP32-S2	Single-core Xtensa LX7	240 MHz	320 KB	Yes	No	Native USB-OTG, low power
ESP32-S3	Dual-core Xtensa LX7	240 MHz	512 KB	Yes	BLE 5	AI vector instructions, camera/LCD support
ESP32-C3	Single-core RISC-V	160 MHz	400 KB	Yes	BLE 5	Low-cost, RISC-V architecture
ESP32-C6	Single-core RISC-V	160 MHz	512 KB	Wi-Fi 6	BLE 5 + Zigbee/Thread	Matter / mesh-ready smart home

SRAM figures include cache/system reserve and vary slightly by SDK version — treat as approximate.

HEAD-TO-HEAD

Arduino Uno vs. ESP32

Arduino Uno

8-bit AVR microcontroller

Cores	1 (8-bit)
Clock	16 MHz
RAM	2 KB
Flash	32 KB
Wireless	None (add-on module needed)
Best for	Simple I/O, learning fundamentals

ESP32

32-bit dual-core SoC

Cores	2 (32-bit)
Clock	240 MHz
RAM	520 KB
Flash	4 MB (typical, external)
Wireless	Wi-Fi + Bluetooth built in
Best for	IoT, wireless sensing, connected robotics

SINGLE-BOARD COMPUTERS

Introduction to Raspberry Pi

The Raspberry Pi is a credit-card-sized single-board computer (SBC) built around a full microprocessor (SoC) — unlike Arduino or ESP32, it boots a complete operating system (typically Raspberry Pi OS, Linux-based) and behaves like a small PC.

- **Runs an OS**

  Boots Linux from an SD card / SSD — multitasking, file system, networking stack.

- **Full CPU**

  ARM Cortex-A application processor, not a simple 8-bit MCU core.

- **Rich I/O**

  USB, HDMI, Ethernet, camera (CSI), display (DSI), plus 40-pin GPIO header.

- **No real-time guarantee**

  An OS scheduler runs many processes — less deterministic timing than a bare MCU.

Where it fits in robotics

Arduino/ESP32 handle fast, low-level control (motors, sensors, PWM). A Raspberry Pi sits above them as the 'brain' — running computer vision, path planning, or a web dashboard — and sends high-level commands down to the microcontroller.

Raspberry Pi Pico (a separate product) IS a microcontroller board (RP2040 chip) — don't confuse it with the Pi 4/5 SBC family.

Raspberry Pi Model Comparison

Model	SoC / CPU	Clock	RAM Options	Key Notes
Pi Zero 2 W	Quad-core Cortex-A53	1.0 GHz	512 MB	Tiny, low-power, Wi-Fi/BLE built in
Pi 3 Model B+	Quad-core Cortex-A53	1.4 GHz	1 GB	Budget SBC, still widely used
Pi 4 Model B	Quad-core Cortex-A72	1.5–1.8 GHz	1 / 2 / 4 / 8 GB	USB 3.0, dual 4K display support
Pi 5	Quad-core Cortex-A76	2.4 GHz	2 / 4 / 8 / 16 GB	2–3x faster than Pi 4, PCIe, RP1 I/O chip
Pi Pico (RP2040)	Dual-core Cortex-M0+	133 MHz	264 KB SRAM	This one is an MCU, not an SBC

Highlighted row: Pico is architecturally a microcontroller, included here only because of naming confusion with the Pi SBC family.

HEAD-TO-HEAD

Arduino vs. ESP32 vs. Raspberry Pi

Attribute	Arduino Uno	ESP32	Raspberry Pi (4/5)
Category	Microcontroller	Microcontroller (SoC)	Single-board computer
Runs	One program (bare metal)	One program (bare metal / RTOS)	Full OS (Linux)
RAM	2 KB	320–520 KB	1–16 GB
Storage	32 KB Flash (code only)	4–16 MB Flash	microSD / SSD (GBs)
Wireless	None built-in	Wi-Fi + Bluetooth	Wi-Fi + Bluetooth
Real-time control	Excellent	Very good	Limited (OS scheduling)
Boot time	Instant	Milliseconds	Seconds (OS boot)
Typical role	Direct sensor/motor control	Wireless sensing & control	High-level processing, vision, AI

DESIGN GUIDE

Choosing the Right Board

1 Choose Arduino when…

- Learning fundamentals of digital/analog I/O
- Direct, precise, real-time control matters
- No wireless connectivity is needed
- Budget and simplicity are priorities

2 Choose ESP32 when…

- The project needs Wi-Fi or Bluetooth
- You still need real-time GPIO/PWM control
- Battery-powered, low-cost IoT node
- WiFi-based sensing (e.g., CSI/HAR) is involved

3 Choose Raspberry Pi when…

- You need to run Python, OpenCV, or an OS
- Camera-based vision or AI inference is needed
- Networking, dashboards, or file storage matter
- It will act as the 'brain' above a microcontroller

CONCLUSION

Key Takeaways

- A microcontroller integrates CPU + RAM + ROM + I/O on one chip to run a single dedicated program.
- Arduino boards differ mainly in memory (Flash/SRAM) and pin count, not raw processing philosophy.
- ESP32 extends the same microcontroller model with wireless connectivity and far more RAM.
- Raspberry Pi is a different category — a full computer (SBC) capable of running an operating system.
- Real embedded/robotics systems often combine layers: Raspberry Pi for high-level logic, ESP32/Arduino for real-time hardware control.